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Course Title: Database System

Course Code: CS-2204

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Lab No: 11

Department Of Software Engineering

DATABASE SYSTEMS LAB REPORT

Part B: Numeric and Date/Time Functions

1. Introduction

Part B of the Scalar SQL Functions lab focuses on **Numeric Functions** and **Date/Time Functions** in MySQL.

Numeric functions are used to perform mathematical operations, rounding, discounts, taxes, and calculations on numerical data.

Date and time functions are used to extract information from dates, calculate differences between dates, format dates, add or subtract time periods, and calculate customer age.

The lab uses the Customer and Product tables from the scalar_lab database.

2. Objectives

The objectives of Part B are:

1. To understand numeric scalar functions.
2. To calculate discounts and taxes.
3. To use ROUND(), FLOOR(), CEIL(), and MOD().
4. To identify odd product IDs.
5. To extract year, month, and day information from dates.
6. To format dates in a readable form.
7. To calculate customer age.
8. To calculate the number of days since joining.
9. To filter records according to a particular year.
10. To identify products launched during a particular quarter.
11. To find customers who joined during the last six months.
12. To calculate product age in days.
13. To add a specific number of days to a date.
14. To combine multiple scalar functions in a single query.

3. Numeric Functions

The main numeric functions used in Part B are:

- ROUND()
- FLOOR()
- CEIL()
- MOD()

These functions are useful for rounding values, finding integer values, and calculating remainders. The lab manual also introduces other numeric functions such as TRUNCATE(), ABS(), POWER(), and SQRT().

4. Date and Time Functions

The main date/time functions used in Part B are:

- YEAR()
- MONTH()
- MONTHNAME()
- DAYNAME()
- DATEDIFF()
- DATE_SUB()
- DATE_ADD()
- TIMESTAMPDIFF()
- DATE_FORMAT()
- CURDATE()

These functions are used to extract date components, calculate differences, format dates, and perform date calculations.

5. Part B — Lab Tasks

Task B1 — Apply a 15% Discount

Objective

Apply a 15% discount to every product and display the discounted price rounded to two decimal places.

SQL Query

Task B1

```
SELECT
   ProdName,
   Price,
   ROUND(Price * 0.85, 2) AS DiscountedPrice
FROM Product;
```

Explanation

A 15% discount means that 85% of the original price remains.

The formula is:

Discounted Price = Price × 0.85

ROUND() is used to display the result with two decimal places.

Result

ProdName	Price	DiscountedPrice
Laptop Pro 15	185000.00	157250.00
Wireless Mouse	2500.00	2125.00
USB-C Cable	800.00	680.00
Office Chair	18500.00	15725.00
Standing Desk	45000.50	38250.43
Notebook A4	350.00	297.50
Ballpoint Pen 10pk	450.00	382.50
Coffee Beans 1kg	1899.99	1614.99
Green Tea Box	650.00	552.50
Bluetooth Speaker	7500.00	6375.00

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Task B2 — Calculate 17% Sales Tax

Objective

Calculate 17% sales tax for every product and calculate the final price including tax.

SQL Query

Task B2

```
SELECT
   ProdName,
   ROUND(Price * 0.17, 2) AS Tax,
   ROUND(Price * 1.17, 2) AS PriceWithTax
FROM Product;
```

Extra options

←T→	ProdName	Tax	PriceWithTax
Edit Copy Delete	Laptop Pro 15	31450.00	216450.00
Edit Copy Delete	Wireless Mouse	425.00	2925.00
Edit Copy Delete	USB-C Cable	136.00	936.00
Edit Copy Delete	Office Chair	3145.00	21645.00
Edit Copy Delete	Standing Desk	7650.09	52650.59
Edit Copy Delete	Notebook A4	59.50	409.50
Edit Copy Delete	Ballpoint Pen 10pk	76.50	526.50
Edit Copy Delete	Coffee Beans 1kg	323.00	2222.99
Edit Copy Delete	Green Tea Box	110.50	760.50
Edit Copy Delete	Bluetooth Speaker	1275.00	8775.00

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Task B3 — FLOOR and CEILING

Objective

Calculate the floor and ceiling of each product's price divided by 1000.

SQL Query

Task B3

```
SELECT
   ProdName,
   Price,
   FLOOR(Price / 1000) AS FloorVal,
   CEIL(Price / 1000) AS CeilVal
FROM Product;
```

Extra options

			ProdName	Price	FloorVal	CeilVal
Edit	Copy	Delete	Laptop Pro 15	185000.00	185	185
Edit	Copy	Delete	Wireless Mouse	2500.00	2	3
Edit	Copy	Delete	USB-C Cable	800.00	0	1
Edit	Copy	Delete	Office Chair	18500.00	18	19
Edit	Copy	Delete	Standing Desk	45000.50	45	46
Edit	Copy	Delete	Notebook A4	350.00	0	1
Edit	Copy	Delete	Ballpoint Pen 10pk	450.00	0	1
Edit	Copy	Delete	Coffee Beans 1kg	1899.99	1	2
Edit	Copy	Delete	Green Tea Box	650.00	0	1
Edit	Copy	Delete	Bluetooth Speaker	7500.00	7	8

Task B4 — Round Price to the Nearest Hundred

Objective

Round each product's price to the nearest hundred.

SQL Query

Task B4

```
SELECT
   ProdName,
   Price,
   ROUND(Price, -2) AS RoundedPrice
FROM Product;
```

Extra options

← T →	ProdName	Price	RoundedPrice
Edit Copy Delete	Laptop Pro 15	185000.00	185000
Edit Copy Delete	Wireless Mouse	2500.00	2500
Edit Copy Delete	USB-C Cable	800.00	800
Edit Copy Delete	Office Chair	18500.00	18500
Edit Copy Delete	Standing Desk	45000.50	45000
Edit Copy Delete	Notebook A4	350.00	400
Edit Copy Delete	Ballpoint Pen 10pk	450.00	500
Edit Copy Delete	Coffee Beans 1kg	1899.99	1900
Edit Copy Delete	Green Tea Box	650.00	700
Edit Copy Delete	Bluetooth Speaker	7500.00	7500

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Task B5 — Products with Odd Product IDs

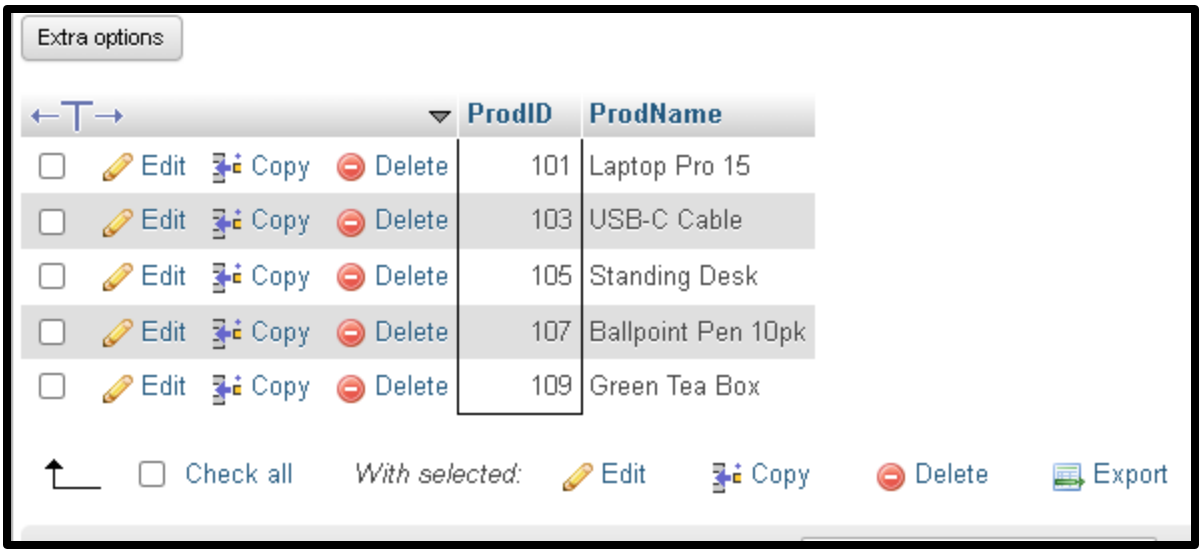
Objective

List products that have an odd `ProdID`.

SQL Query

Task B5

```
SELECT
   ProdID,
   ProdName
FROM Product
WHERE MOD(ProdID, 2) = 1;
```



Task B6 — Customer Joining Date Information

Objective

For each customer, display the year, month name, and day of the week on which they joined.

SQL Query

Task B6

```
SELECT
   CustName,
   JoinDate,
   YEAR(JoinDate) AS JoinYear,
   MONTHNAME(JoinDate) AS JoinMonth,
   DAYNAME(JoinDate) AS JoinDay
FROM Customer;
```

Extra options

			CustName	JoinDate	JoinYear	JoinMonth	JoinDay
Edit	Copy	Delete	Ali Khan	2022-01-15	2022	January	Saturday
Edit	Copy	Delete	Sara Iqbal	2022-04-22	2022	April	Friday
Edit	Copy	Delete	HAMZA RAZA	2023-02-10	2023	February	Friday
Edit	Copy	Delete	Ayesha Noor	2023-05-18	2023	May	Thursday
Edit	Copy	Delete	bilal ahmed	2023-09-01	2023	September	Friday
Edit	Copy	Delete	Fatima Sheikh	2024-01-12	2024	January	Friday
Edit	Copy	Delete	Usman Tariq	2024-06-30	2024	June	Sunday
Edit	Copy	Delete	Maira Javed	2024-08-25	2024	August	Sunday

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Task B7 — Format Date of Birth

Objective

Display each customer's date of birth in the format:

```
DD-Month-YYYY
```

SQL Query

Task B7

```
SELECT
   CustName,
   DOB,
   DATE_FORMAT(DOB, '%d-%M-%Y') AS FormattedDOB
FROM Customer;
```

Extra options

			CustName	DOB	FormattedDOB
Edit	Copy	Delete	Ali Khan	1995-04-12	12-April-1995
Edit	Copy	Delete	Sara Iqbal	1998-11-20	20-November-1998
Edit	Copy	Delete	HAMZA RAZA	1997-08-05	05-August-1997
Edit	Copy	Delete	Ayesha Noor	1999-02-14	14-February-1999
Edit	Copy	Delete	bilal ahmed	2000-06-30	30-June-2000
Edit	Copy	Delete	Fatima Sheikh	1996-10-25	25-October-1996
Edit	Copy	Delete	Usman Tariq	2001-03-18	18-March-2001
Edit	Copy	Delete	Maira Javed	1994-12-09	09-December-1994

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Task B8 — Calculate Current Age

Objective

Calculate each customer's current age in years.

SQL Query

Task B8

```
SELECT
   CustName,
   DOB,
   TIMESTAMPDIFF(YEAR, DOB, CURDATE()) AS Age
FROM Customer;
```

Extra options

			CustName	DOB	Age
Edit	Copy	Delete	Ali Khan	1995-04-12	31
Edit	Copy	Delete	Sara Iqbal	1998-11-20	27
Edit	Copy	Delete	HAMZA RAZA	1997-08-05	29
Edit	Copy	Delete	Ayesha Noor	1999-02-14	27
Edit	Copy	Delete	bilal ahmed	2000-06-30	26
Edit	Copy	Delete	Fatima Sheikh	1996-10-25	29
Edit	Copy	Delete	Usman Tariq	2001-03-18	25
Edit	Copy	Delete	Maira Javed	1994-12-09	31

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Task B9 — Days Since Joining

Objective

Calculate how many days have passed since each customer joined.

SQL Query

Task B9

```
SELECT
   CustName,
   JoinDate,
   DATEDIFF(CURDATE(), JoinDate) AS DaysSinceJoin
FROM Customer;
```

Extra options

				CustName	JoinDate	DaysSinceJoin
☐	Edit	Copy	Delete	Ali Khan	2022-01-15	1709
☐	Edit	Copy	Delete	Sara Iqbal	2022-04-22	1612
☐	Edit	Copy	Delete	HAMZA RAZA	2023-02-10	1318
☐	Edit	Copy	Delete	Ayesha Noor	2023-05-18	1221
☐	Edit	Copy	Delete	bilal ahmed	2023-09-01	1115
☐	Edit	Copy	Delete	Fatima Sheikh	2024-01-12	982
☐	Edit	Copy	Delete	Usman Tariq	2024-06-30	812
☐	Edit	Copy	Delete	Maira Javed	2024-08-25	756

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Task B10 — Customers Who Joined in 2023

Objective

List customers who joined during the year 2023.

The task requires using a date function rather than `BETWEEN`.

SQL Query

Task B10

```
SELECT
   CustID,
   CustName,
   JoinDate
FROM Customer
WHERE YEAR(JoinDate) = 2023;
```

Extra options

	CustID	CustName	JoinDate
Edit Copy Delete	3	HAMZA RAZA	2023-02-10
Edit Copy Delete	4	Ayesha Noor	2023-05-18
Edit Copy Delete	5	bilal ahmed	2023-09-01

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Task B11 — Products Launched in Q4

Objective

List products launched in any year's fourth quarter.

The fourth quarter consists of:

- October
- November
- December

SQL Query

Task B11

```
SELECT
   ProdID,
   ProdName,
   LaunchDate
FROM Product
WHERE MONTH(LaunchDate) IN (10, 11, 12);
```

Extra options

	ProdID	ProdName	LaunchDate
Edit Copy Delete	103	USB-C Cable	2021-11-05
Edit Copy Delete	109	Green Tea Box	2022-12-12

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Explanation

`MONTH()` extracts the month number from the launch date.

The query selects products where the month is:

`10, 11, or`

Task B12 — Customers Who Joined Within Last 6 Months

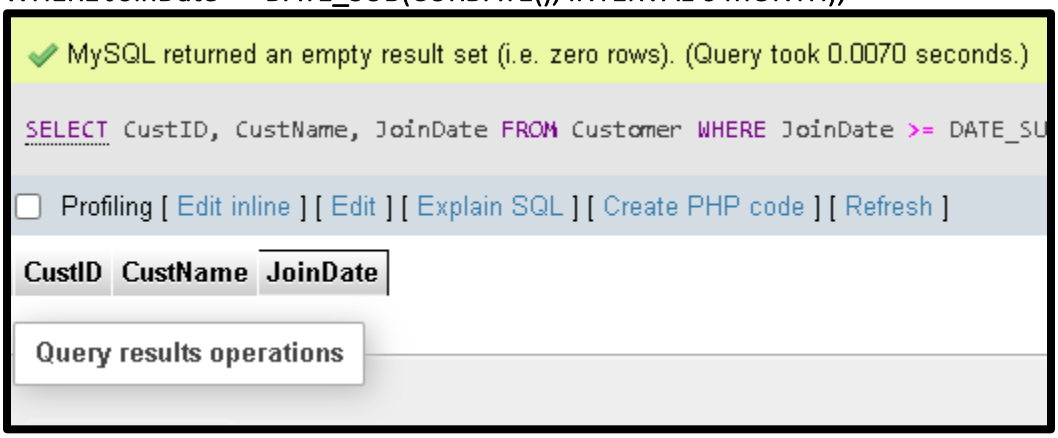
Objective

Find customers who joined within the last six months from today.

SQL Query

Task B12

```
SELECT
   CustID,
   CustName,
   JoinDate
FROM Customer
WHERE JoinDate >= DATE_SUB(CURDATE(), INTERVAL 6 MONTH);
```



Explanation

`CURDATE()` returns today's date.

`DATE_SUB()` subtracts six months from today's date.

The query then selects customers whose joining date is on or after that calculated date.

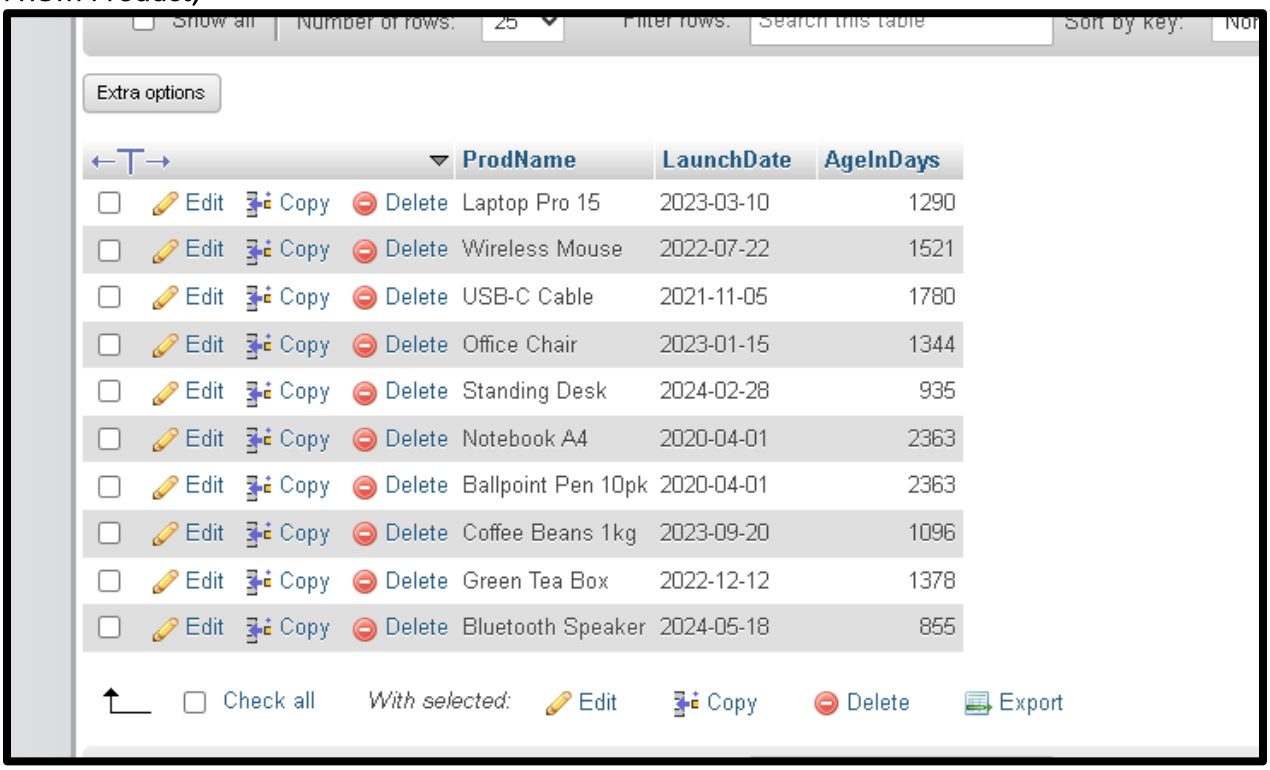
Task B13 — Product Age in Days

Objective

Calculate how many days have passed since each product was launched.

SQL Query

```
Task B13
SELECT
   ProdName,
   LaunchDate,
   DATEDIFF(CURDATE(), LaunchDate) AS AgeInDays
FROM Product;
```



Extra options

← T →	ProdName	LaunchDate	AgeInDays
Edit Copy Delete	Laptop Pro 15	2023-03-10	1290
Edit Copy Delete	Wireless Mouse	2022-07-22	1521
Edit Copy Delete	USB-C Cable	2021-11-05	1780
Edit Copy Delete	Office Chair	2023-01-15	1344
Edit Copy Delete	Standing Desk	2024-02-28	935
Edit Copy Delete	Notebook A4	2020-04-01	2363
Edit Copy Delete	Ballpoint Pen 10pk	2020-04-01	2363
Edit Copy Delete	Coffee Beans 1kg	2023-09-20	1096
Edit Copy Delete	Green Tea Box	2022-12-12	1378
Edit Copy Delete	Bluetooth Speaker	2024-05-18	855

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Explanation

`DATEDIFF()` calculates the number of days between today and the product's launch dat

Task B14 — Date 90 Days After Launch

Objective

Calculate the date exactly 90 days after each product's launch date.

SQL Query

Task B14

```
SELECT
   ProdName,
   LaunchDate,
   DATE_ADD(LaunchDate, INTERVAL 90 DAY) AS NinetyDaysLater
FROM Product;
```

Extra options

← T →	ProdName	LaunchDate	NinetyDaysLater
Edit Copy Delete	Laptop Pro 15	2023-03-10	2023-06-08
Edit Copy Delete	Wireless Mouse	2022-07-22	2022-10-20
Edit Copy Delete	USB-C Cable	2021-11-05	2022-02-03
Edit Copy Delete	Office Chair	2023-01-15	2023-04-15
Edit Copy Delete	Standing Desk	2024-02-28	2024-05-28
Edit Copy Delete	Notebook A4	2020-04-01	2020-06-30
Edit Copy Delete	Ballpoint Pen 10pk	2020-04-01	2020-06-30
Edit Copy Delete	Coffee Beans 1kg	2023-09-20	2023-12-19
Edit Copy Delete	Green Tea Box	2022-12-12	2023-03-12
Edit Copy Delete	Bluetooth Speaker	2024-05-18	2024-08-16

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Explanation

`DATE_ADD()` adds a specified time interval to a date.

Here, 90 days are added to every product's launch date.

Task B15 — Combined Challenge

Objective

Create a single column called Summary containing customer information in the following format:

```
Hello ALI KHAN, age 31, joined Jan 2022
```

SQL Query

Task B15

```
SELECT
   CONCAT(
      'Hello ',
      UPPER(TRIM(CustName)),
      ', age ',
      TIMESTAMPDIFF(YEAR, DOB, CURDATE()),
      ', joined ',
      DATE_FORMAT(JoinDate, '%b %Y')
   ) AS Summary
FROM Customer;
```

Functions Used

This task combines multiple scalar functions:

- `TRIM()` — removes unnecessary spaces.
- `UPPER()` — converts the name to uppercase.
- `TIMESTAMPDIFF()` — calculates age.
- `DATE_FORMAT()` — formats the joining date.
- `CONCAT()` — combines everything into one sentence.

Example

```
Hello ALI KHAN, age 31, joined Jan 2022
```

6. Important Date Format Specifiers

The lab uses several DATE_FORMAT() specifiers.

Specifier	Meaning
`%Y`	Four-digit year
`%y`	Two-digit year
`%m`	Month number
`%M`	Full month name
`%b`	Short month name
`%d`	Day of month
`%D`	Day with suffix
`%W`	Full weekday name
`%a`	Short weekday name

For example:

```
DATE_FORMAT(JoinDate, '%d-%M-%Y')
```

produces a date such as:

```
15-January-2022
```

The manual provides the `DATE_FORMAT` specifier table for this purpose.

7. Conclusion

In Part B of the Scalar SQL Functions lab, numeric and date/time functions were successfully applied to the Customer and Product tables.

Numeric functions were used to calculate discounts, taxes, rounded values, floor and ceiling values, and odd product IDs. Date/time functions were used to extract date components, format dates, calculate customer ages, calculate the number of days between dates, identify records from specific time periods, and add or subtract time intervals.

The final combined challenge demonstrated how several scalar functions can be nested and combined to transform database records into meaningful and human-readable information